Metacognition and Self-Regulated Learning

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Relevant textbook section: Chapters 9 and 11, sections on metacognition, learning strategies, and self-regulated learning

Metacognitive knowledge and regulation

Metacognition is often described as thinking about thinking, but it includes more than awareness of one's own mental activity. Metacognitive knowledge concerns what individuals understand about themselves as learners, about the requirements of tasks, and about the characteristics and possible uses of strategies. Knowledge about a strategy includes both how it can be carried out and the conditions under which it is likely to be useful. Metacognitive regulation concerns the actions through which learners direct their cognitive activity. Planning involves setting purposes, anticipating demands, allocating resources, and choosing an initial approach. Monitoring involves checking comprehension and progress while activity unfolds. Evaluation concerns judgments about outcomes and the effectiveness of prior decisions. These components interact, and regulation may be revised repeatedly during a task rather than proceeding in a fixed linear sequence.

Strategies and strategic knowledge

Learning strategies are intentional procedures for improving understanding, remembering information, or completing learning activities. Some strategies support rehearsal and retrieval, whereas others emphasize elaboration, organization, comparison, or the construction of relationships among ideas. Their usefulness depends on what must be learned and what kind of performance is expected. For this reason, knowledge of many strategies does not necessarily imply effective strategic learning. Learners also need conditional knowledge: an understanding of when and why a strategy is appropriate. Strategic activity entails selecting an approach in view of a goal, judging whether it is productive, and changing it when needed. Persisting with the same approach may sometimes reflect productive effort, but greater time or repetition alone does not establish that learning is improving. An effective strategy is evaluated with reference to the demands and standards of a particular task.

Self-regulated learning

Self-regulated learning describes the active coordination of thoughts, motivation, behavior, and environmental resources toward learning goals. It includes cognitive and metacognitive processes, but it is not reducible to either of them. Motivational beliefs influence whether learners initiate and sustain regulation, while behavioral and contextual decisions influence the conditions in which cognition takes place. Regulation can include managing attention, setting proximal goals, controlling distractions, seeking information, requesting assistance, and using feedback. Self-regulated learning should not be equated with studying in isolation. Interaction with teachers, peers, and instructional resources can provide information and support that learners use in regulating their work. Over time, learners may develop greater independence in making these decisions, although their regulation remains responsive to the situations and tasks they encounter.

Monitoring, calibration, and adaptation

A distinctive challenge of regulation is that subjective impressions of understanding do not always match demonstrated performance. Monitoring therefore requires evidence against which progress can be judged, rather than reliance solely on how familiar material feels. Feedback, self-questioning, criteria for performance, and attempts to retrieve or explain knowledge can all inform judgments about progress. The correspondence between a learner's judgment and actual performance is sometimes described as calibration. Monitoring is consequential when it informs control: for example, deciding to allocate further attention, alter the approach, or reconsider a prior judgment. Evaluation after an activity may also shape planning in a later one. Thus, metacognitive knowledge, regulatory actions, and self-regulated learning are closely related but analytically distinct. Explaining learning outcomes requires attention not only to whether strategies were used but also to how decisions about strategies were made and revised. Regulation also varies with the learner’s understanding of the task itself. A vague or inaccurate representation of the goal can lead to strategies that are carried out consistently yet serve an inappropriate purpose. Learners may require support to interpret performance criteria and translate broad aims into manageable intermediate goals. As a result, regulation should be assessed through relationships among task understanding, choice of strategy, evidence of progress, and adjustment. Individual differences in experience and knowledge may affect these processes, as may the clarity and structure of the learning environment. Self-regulation is best understood as developing and context-sensitive rather than as a fixed personal trait.